The Brain as a Plastic Observable-Algebra Network
The Conditional Fixed-Point Workspace $\mathcal{W}_{\text{int}}$

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Abstract

The self-model is often treated as a point-like subject, while the brain is often described as an information-processing machine. This paper is a standalone anchoring interface paper: it gives a structural interface between an observable-algebra framework and one biological observer regime, without extending the Foundation formal theory and without making an empirical neuroscience claim.

The core proposal is that a brain-centered embodied observer regime can anchor an accessible algebra $\mathcal{A}_c$ at biological scale. Neural read-out structures are read as idealized local subalgebras $\mathcal{B}_i$, synaptic and functional couplings are modeled by normal unital completely positive maps $\Phi_{ij}(t)$, coupling history is represented by $K(t,s)$, and the reportable self-model is read as the internal fixed-point workspace $\mathcal{W}_{\text{int}}$. In the Formal Core, $\mathcal{W}_{\text{int}}$ is the exact read-out equalizer and is a von Neumann subalgebra only under the stated finite or compact automorphic group action; the positive-δ tolerance locus is not asserted to be an algebra.

All claims are conditional and interface-level. Neuroscience examples are used only as compatible re-readings, not as evidence for the framework. The finite-dimensional Pauli-qubit witness (Proposition 6.5) verifies a non-trivial $\mathcal{W}_{\text{int}}$ and non-zero private factors in one Type I model, while the general private-complement condition (P) remains open. The object $\mathcal{W}_{\text{int}}$ is type-distinct from H_{st} and H_{12}; Foundation II supplies analogy and motivation only. The paper does not prove qualia, does not derive an empirically testable construction of $\mathcal{B}_{\text{brain}}$, and does not authorize consciousness migration, digital immortality, or first-person identity transfer.

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1 Introduction: The Need for a Biological Anchor

The Genesis and Consciousness papers describe consciousness-adjacent structure at the level of accessible algebras, shared sectors, modular flow, source-level feedback, and self-description limits. Those papers do not, by themselves, specify a concrete biological observer regime. A reader can therefore ask a simple anchoring question: if an accessible algebra $\mathcal{A}_c$ is not a free-floating abstraction, how is it anchored in the biological case?

This paper answers that question only at the level of a framework interface. It does not introduce Foundation IV. It does not extend the formal Foundation series. It also does not present an empirical neuroscience model with estimated parameters, data fits, or clinical predictions.

The proposed interface is that the brain-centered embodied observer regime can be read as a biological anchor for an accessible algebra. In that regime, the brain is a central plastic algebraic organ, but it is not the whole observer regime. The body, sensorimotor loops, interoception, affective systems, environmental coupling, tools, records, and other agents also constrain what is accessible and reportable.

The brain is not the source of consciousness in a reductive sense, nor a container of an inner world. Consciousness, in this framework, is not a thing that requires a source, but a structural modality of an accessible algebra anchored through a biological observer regime. The brain is such a central anchoring component: it helps stabilize a plastic internal integration workspace $\mathcal{W}_{\text{int}}$ that can later be reported as "I."

The guiding contrast is therefore not between a ghostly subject and a physical brain. The guiding contrast is between a many-channel, plastic, history-dependent observer regime and the later reportable self-model that this regime stabilizes. The reportable "I" is not discarded as unreal. It is read as an effective stable structure within $\mathcal{A}_c$, not as an independent point-subject.

Contribution

The paper makes three contributions. First, it gives a self-contained biological anchoring interface for the observable-algebra framework. A reader need not accept any empirical claim about the brain to understand the interface: the key question is how an accessible algebra can be anchored through an embodied observer regime.

Second, the paper turns the internal self-model into a formal target. Instead of treating the "I" as either a primitive subject or a mere illusion, it asks what kind of stable readable substructure could be supported by multiple internal read-out structures. The answer is conditional: under a selected coarse-graining, specified conditional expectations, specified read-out comparison maps, and specified coupling maps, the exact operator equalizer defines an internal integration workspace $\mathcal{W}_{\text{int}}$. A finite-dimensional witness shows that the formal conditions are jointly satisfiable (Propositions 6.1–6.5).

Third, the paper separates the algebraic contribution from the neuroscience interface. Neuroscience examples are used to show where the language might touch existing phenomena, not to validate the framework. This separation is essential. Without it, the paper would risk borrowing neuroscience authority for a formal structure that has not been derived from data. With it, the paper can be read as an algebra-first interface proposal whose empirical relevance remains a later research program.

How to read the structural claims

Most claims in the paper have the form "X can be read as Y" rather than "$X = Y$." This is not rhetorical caution alone. It marks the level of the paper. The biological phenomenon is not being reduced to an algebraic object, and the algebraic object is not being inferred from the biological phenomenon. The paper proposes a partial structural reading between the two.

This also explains why the paper uses conditional propositions rather than unconditional theorems. The propositions are not claims that brains automatically satisfy the hypotheses. They say that, if an embodied observer regime is idealized by the specified operator-algebraic datum, then the internal workspace and common/private decomposition are well-defined within that datum. The formal burden is therefore explicit rather than hidden.

What this paper does not do

- It does not solve the hard problem.
- It does not prove that the brain produces consciousness.
- It does not propose that consciousness has a source requiring identification.
- It does not derive an empirically testable construction of $\mathcal{B}_{\text{brain}}$ from neuroscience data.
- It does not replace Global Workspace Theory, predictive processing, higher-order theories, or clinical neuroscience.
- It does not authorize consciousness migration, digital immortality, or first-person identity transfer.

Plan of the paper

Section 2 gives the minimal self-contained observable-algebra vocabulary needed for the paper. Section 3 explains why accessible algebras require embodied observer regimes. Section 4 formulates the brain-level reading as a plastic observable-algebra network. Section 5 describes the self-model as an internal integration workspace.

Section 6 gives the formal core: a conditional hull construction, a conditional common/private decomposition, and a fragmentation criterion. Section 7 gives a limited compatible re-reading of first-person report timing. Section 8 compresses the neuroscience interfaces into a table and selected discussions. Section 9 separates guardrails from limitations. Section 10 states a research program, and Section 11 restates the core anchor claim. Appendix A gives the fuller formal skeleton.

2 A Minimal Self-Contained Framework: Imported Vocabulary

This section gives the imported minimal vocabulary required for the present paper. The terms are compatible with Foundation I, Foundation II, and Foundation III of the Consciousness Series [1, 2, 3], but the definitions below are intended to be readable without consulting those papers. Nothing in this section is meant to re-prove the Foundation results, and no Foundation theorem is imported into the formal core below.

Definition 2.1 (Observable algebra)**.** An observable algebra $\mathcal{A}$ is an algebraic language for quantities that can be read, stabilized, transformed, or reported by a specified regime. In this paper $\mathcal{A}$ is idealized as a von Neumann algebra. The von Neumann setting is used because it makes states, normal maps, conditional expectations, and weak operator closure available in one common language. This is a structural idealization, not a claim that neural tissue literally presents a naturally given von Neumann algebra.

The word "observable" should be read structurally rather than psychologically. An observable is not necessarily an item that is consciously noticed. It is an element of the formal language through which a regime can, in principle, stabilize distinctions, update states, or support later report. This distinction matters because the paper is not trying to identify conscious contents one by one with algebra elements. It is specifying the structural level at which access, readability, and report can be discussed.

The von Neumann idealization is deliberately stronger than what the neuroscience interface can presently justify. It is used because the formal questions of this paper require closure, states, and conditional expectations. A weaker C^*-algebraic language would be possible in principle, but the present construction repeatedly refers to normal states and σ-weak closure. For that reason, the von Neumann setting is the cleanest operator-algebraic envelope for a first standalone interface paper.

Definition 2.2 (Ambient algebra)**.** The ambient algebra $\mathcal{A}_U$ denotes the background observable algebra relative to which accessible substructures are selected. It is not treated here as a collection of already separated objects. It is the formal substrate from which finite regimes induce accessible sectors.

The ambient algebra is not an empirically specified object in this paper. It is a bookkeeping device for the fact that access is partial. The biological observer regime does not encounter an already segmented world and then add subjective labels to it. Rather, the formal picture starts from a broader observable context and asks which substructure is available under the constraints of a finite regime.

Definition 2.3 (Accessible algebra)**.** An accessible algebra $\mathcal{A}_c \subseteq \mathcal{A}_U$ is the observable algebra available to a finite observer regime. It is not an arbitrary psychological viewpoint. It is the algebraic expression of what can be read, stabilized, updated, reported, and acted through under the constraints of that regime.

This is the main self-contained hinge of the paper. An accessible algebra is not a mind, a brain region, or a representation stored in a head. It is the formal expression of a regime's accessible observables. For the biological case, this means that $\mathcal{A}_c$ is anchored through the brain-centered embodied regime, but it is not identical with the brain. The brain is a central anchoring component, while $\mathcal{A}_c$ is the algebraic object whose access conditions are constrained by the whole regime.

Definition 2.4 (Observer regime)**.** An observer regime is the physical, biological, and historical set of conditions through which an accessible algebra is anchored. An accessible algebra only has operational meaning relative to such a regime. In the biological case considered here, the regime is brain-centered but embodied: body, sensorimotor loops, interoception, environmental coupling, tools, records, and social interaction all constrain access.

The observer regime is the paper's substitute for a point-like subject. It is also the paper's guard against a brain-in-a-vat reading. A finite observer regime has sensors, effectors, metabolic

constraints, bodily states, memory traces, and environmental couplings. These conditions do not merely deliver inputs to a detached algebra. They help determine which observables can be stabilized and which distinctions can become reportable.

This regime language is also meant to avoid a common ambiguity in consciousness theory. A theory may speak as if there is a subject to whom contents appear, and then ask how the brain generates those contents. The present framework reverses that order. It first specifies the conditions under which an accessible algebra is available. Only then does it ask how a reportable self-model can stabilize inside that accessible algebra. The subject is not inserted as a primitive term at the beginning.

The approach therefore does not deny first-person structure. It denies that first-person structure should be treated as an unanalysed point. The reportable first person is a late and effective structure within a regime. It has practical and ethical force because the regime can plan, remember, repair, and be trained. But its force does not require it to be an independent owner of the algebra.

Definition 2.5 (Typed hull namespace)**.** Three objects previously described with overlapping "stable hull" language are kept type-distinct. H_{st} is Foundation I's source-algebraic stable-hull construction, H_{12} is Foundation II's external shared accessible sector for independent source states [2], and $\mathcal{W}_{\text{int}}$ is the internal cognitive integration workspace studied in this paper. No equality, functor, or theorem-level inheritance among these three objects is asserted.

The external sector H_{12} and the internal workspace $\mathcal{W}_{\text{int}}$ should not be confused. The former concerns agreement between independently supplied source states. The latter concerns agreement among read-out comparison maps inside one accessible algebra and must solve its own non-vacuity problem. Foundation II is therefore an analogy and source of motivation only; its theorem does not establish the construction used here.

Definition 2.6 (Private complement)**.** The private complement is the part of a regime or read-out channel that does not enter the shared-readable sector. In external settings it marks what does not become public overlap. In the internal setting considered here, it marks what remains local to a bodily, affective, memory, or report-limited channel.

The private complement is not a hidden metaphysical substance. It is a formal way of saying that not everything shaping a reportable self-model is itself shared-readable across the relevant channels. A bodily state may bias report without becoming linguistic. An affective salience signal may constrain action without becoming a public reason. The private complement records this asymmetry without treating it as a separate subject.

Definition 2.7 (History-dependent coupling)**.** The symbol $K(t, s)$ denotes a history-dependent coupling object. It is not a single neural kernel derived from data. It is a structural placeholder for non-Markovian dependence: past interactions change which future states are easier to read, synchronize, report, or act through.

The notation $K(t, s)$ is intentionally generic. It can be read as a history-dependent kernel, a coupling-history object, or a record of how prior interactions constrain later synchronization. At the biological interface, synaptic plasticity, network reconfiguration, habit formation, and affective conditioning may all provide partial empirical interfaces to this role. The paper does not identify $K(t, s)$ with any one of them.

The same caution applies to $\Phi_{ij}(t)$. A coupling map is not a wire carrying a pre-existing message from one component to another. It is a structural representation of how one read-out

structure can change the future readability or stabilization conditions of another. In neural language, such coupling may be mediated by synapses, recurrent loops, neuromodulation, functional connectivity, or body-mediated feedback. In the present paper these are not collapsed into one biophysical mechanism. They are placed under one formal heading only at the level of structural readability.

The point of the present paper is not to identify the brain with $\mathcal{A}_c$. The point is to describe how a brain-centered embodied regime can anchor one biological instance of an accessible algebra. In that reading, $\mathcal{A}_c$ remains the algebraic object, while the biological regime is the anchoring condition.

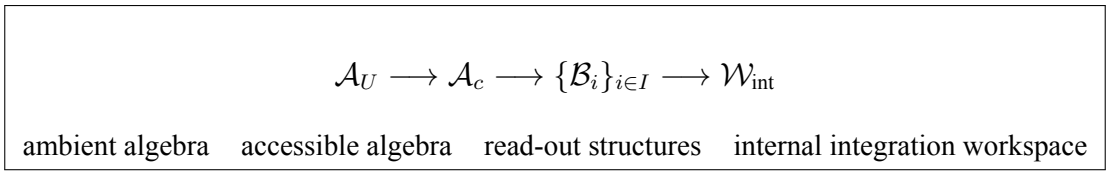


Figure 1: Schematic only. The figure shows the minimal framework used in this paper: $\mathcal{A}_U$, $\mathcal{A}_c$, internal read-out structures $\{\mathcal{B}_i\}$, and the internal integration workspace $\mathcal{W}_{\text{int}}$.

2.1 Translation Conventions

Because the paper is an interface paper, several terms have two readings: an operator-algebraic reading and a biological or phenomenological interface reading. The following table fixes the translation conventions used throughout the paper.

Term	Formal role	Interface role
$\mathcal{A}_U$	Ambient observable algebra.	Background structural context, not an empirically measured universe.
$\mathcal{A}_c$	Accessible von Neumann subalgebra.	What a finite embodied regime can read, stabilize, update, and report.
$\mathcal{B}_i$	Idealized read-out subalgebra at a fixed scale.	Neural, bodily, memory, affective, or linguistic channel read as a local structure.
E_i	Faithful normal conditional expectation onto $\mathcal{B}_i$.	Idealized selection of channel-readable content.
R_α	Normal unital completely positive comparison map.	Idealized test of exact or δ-approximate internal agreement.
$\Phi_{ij}^K(t)$	Normal unital completely positive coupling map.	Plastic coupling between read-out structures.
$K(t,s)$	History-dependent coupling object.	Memory, habit, plasticity, or prior coupling history at structural level.
(δ, T)	Operator-norm tolerance and report-window length.	Declared modeling parameters, not quantities inferred from neuroscience.
$\mathcal{W}_{\text{int}}$	Exact fixed-point equalizer under the Formal Core hypotheses.	Stable shared-readable self-model, later reportable as "I."

Table 1: Translation conventions. The biological column gives compatible interface readings only; it does not identify the biological phenomenon with the formal object.

The table is also a warning against category mistakes. A cortical network is not literally a von Neumann algebra; it is read through one when the paper asks about observable access and reportability. A synapse is not literally a completely positive map; synaptic and functional coupling can be modeled by such maps at a selected structural level. The value of the formalism lies in making the conditions of shared readability explicit, not in replacing biology with algebra.

2.2 Why This Minimal Vocabulary Is Enough

The paper does not require the full machinery of the Foundation series. It uses only four structural ideas. First, access is partial and regime-dependent. Second, stable overlap can be studied algebraically. Third, not everything local to a channel becomes shared-readable. Fourth, history changes future readability. These four ideas are enough to formulate the brain-level anchor without importing the full source chain, modular-flow, or perfuming-loop machinery.

This is why the paper can be standalone. Foundation I–III provide the background motivation and a broader research program, but the argument below only needs the self-contained vocabulary introduced here. A reader who accepts the definitions in this section can follow the later construction even without accepting the whole Consciousness Series.

3 Accessible Algebras Require Embodied Observer Regimes

An accessible algebra is specified relative to an observer regime. It is not a free-floating abstraction. What can be read, updated, reported, and acted through depends on the embodied conditions under which accessibility is realized.

For the biological case, the observer regime is brain-centered but not brain-only. The brain is the central plastic algebraic organ of the regime. The full regime also includes the body, sensorimotor loops, interoceptive and affective systems, environmental coupling, tools, records, habits, and social interaction.

This distinction prevents a common drift. The paper does not claim that the brain is $\mathcal{A}_c$. It claims that the brain-centered embodied regime anchors an accessible algebra. The accessible algebra is constrained by the whole embodied regime, even when the brain provides the most plastic and integrative component.

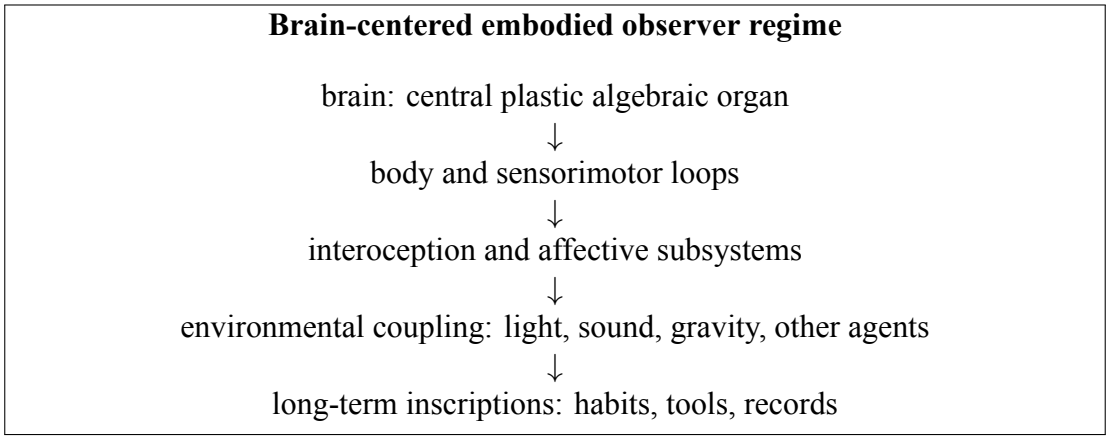


Figure 2: Schematic only. The brain is central in the biological anchor, but it is not the whole observer regime.

The embodied regime also prevents a second drift, namely the brain-in-a-vat reading. The accessible algebra is not anchored by an isolated neural object floating apart from sensorimotor and environmental constraints. It is anchored by a coupled regime whose neural component is central but not sufficient by itself.

This paper therefore treats the brain as an anchoring component rather than as a container of a world. The world as read by the regime is not inside the brain. The readable conditions under which a world is stabilized are constrained through the regime.

This regime-level language also explains why the paper does not start from an isolated computational brain. A purely neural description may be appropriate for many empirical purposes, but the accessible algebra of a biological observer is constrained by more than neural tissue. Posture, interoceptive load, sensorimotor contingencies, sleep history, tools, written records, and social interaction can change what is readable and reportable. These factors do not have to be modeled in detail here, but they justify treating the brain as central within an embodied regime rather than as the whole regime.

The point is not to dilute the role of the brain. The brain remains the dominant plastic integration organ of the biological anchor. The point is to avoid making the algebraic object $\mathcal{A}_c$ identical to a neural object before the regime of access has been specified. In the notation used later, a candidate brain algebra $\mathcal{B}_{\text{brain}}$ may be a crucial substructure of $\mathcal{A}_c$, but the embodied regime also constrains the state, read-out maps, report windows, and environmental couplings.

4 The Brain as a Plastic Observable-Algebra Network

The brain can be read, at a chosen biological coarse-graining scale, as a plastic observable-algebra network. This is a structural reading, not an empirical derivation of a neural algebra from data. The reading is useful because it allows neural plasticity, coupling, memory, and self-model stabilization to be discussed in the same language.

Two overstatements must be ruled out at the start. A neural read-out unit is not a quantum state. A synapse is not literal quantum entanglement. Both claims are too strong, and both would push the paper toward a quantum-brain hypothesis that is not being made.

Instead, choose a biological coarse-graining scale. At that scale, a read-out unit may be a single neuron, a neural ensemble, a cortical column, a functional network, a sensorimotor loop, a memory subsystem, an affective subsystem, or a language subsystem. We denote the corresponding effective read-out subalgebras by $\{\mathcal{B}_i\}_{i\in I}$, with $\mathcal{B}_i \subseteq \mathcal{A}_c$.

At the biological interface, a neural read-out unit is not a passive node but a local integrator. Dendritic inputs, excitatory and inhibitory modulation, and thresholded action-potential generation already implement a local read-out operation before network-level coupling is considered [5]. Network-level coupling $\Phi_{ij}(t)$ then composes such local integrations into larger patterns.

The objects $\{\mathcal{B}_i\}_{i\in I}$ are called effective read-out structures in the main text. In the idealized setting of Appendix A they are assumed to be von Neumann subalgebras of $\mathcal{A}_c$. At the biological interface this idealization is a modeling choice, not a claim that every neural ensemble is literally closed under multiplication, adjoint, and operator-topology limits.

The choice of coarse-graining is not a cosmetic detail. At a single neuron scale, one might emphasize dendritic integration, spike thresholds, and local input-output selectivity. At an ensemble scale, one might emphasize synchrony, oscillatory coordination, and functional connectivity. At a behavioral subsystem scale, one might emphasize sensorimotor affordances, affective salience, memory retrieval, or language report. The notation $\mathcal{B}_i$ is deliberately flexible enough to cover these possibilities, but the formal construction in Section 6 always fixes one scale before making a statement.

This prevents a common false precision. The paper does not claim that there are exactly n biologically privileged subalgebras, nor that a single neuron, cortical column, and language subsystem all belong to the same clean algebraic level. It says that, once a scale is selected, the relevant read-out structures can be idealized as subalgebras for the purpose of asking whether a stable internal overlap can be constructed.

Scale	**Possible read-out structure**	**Reason for caution**
Cellular	Single neuron or local dendritic integration.	Too fine-grained for report-level self-modeling by itself.
Mesoscopic	Neural ensemble, column, or recurrent local circuit.	May support stable patterns, but boundaries are model-dependent.
Network	Functional network or large-scale integration pattern.	Closer to report and action, but farther from cellular mechanism.
Embodied subsystem	Sensorimotor loop, affective subsystem, memory system, language system.	Useful for self-model analysis, but mathematically idealized as a subalgebra.

Table 2: Possible coarse-graining scales for $\mathcal{B}_i$. The formal construction fixes one scale; it does not assert invariance across all scales.

Biological structure	**Structural reading**
Neural read-out unit i	Local effective read-out subalgebra $\mathcal{B}_i$ at a chosen biological coarse-graining scale.
Synaptic or functional coupling ij	Plastic coupling channel or algebraic edge $\Phi_{ij}(t)$.
Plasticity	History-dependent modification of coupling channels.
Memory	Coupling history, not a static storage file.
Learning	Modification of future readability, synchronization, and action-candidate stabilization.
Self-model	Network-stabilized internal integration workspace $\mathcal{W}_{\text{int}}$.

Table 3: A structural reading, not an equality table. Each entry is to be read as "can be modeled as" at a chosen coarse-graining scale.

The key shift is that a coupling is not read as an information wire. The wire metaphor is useful in engineering contexts, but it suggests that pre-existing information objects travel along edges. The algebraic-edge reading is different: a coupling channel changes which future states can be jointly stabilized and read.

At the neural interface, this means that synaptic coupling is better read as a plastic electrochemical interface than as a fixed information line. Chemical synapses convert electrical activity into transmitter release and then back into postsynaptic electrical change; the structural point is not the biochemical detail itself, but the fact that coupling is state-dependent, modulatory, and plastic [6, 7]. This supports the use of $\Phi_{ij}(t)$ as a coupling-channel notation rather than a wire notation.

In this sense, a synaptic or functional coupling is a plastic edge in a network of local read-out algebras. Each experience modifies the future conditions of readability. It changes which channels synchronize, which states become easier to stabilize, and which action candidates become more likely to enter reportable dynamics.

Why not a standard neural network model? Standard neural network models can describe input-output mappings, learned weights, prediction errors, and task performance. The present paper is addressing a different structural layer: the algebraic conditions of observable access, the conditions under which states are readable, the distinction between shared and private sectors, the location of the report boundary, and the role of history-dependent coupling. This is not a competing neural network model. Rather, a neural network model can be read as one possible concrete coupling instance inside a broader $\Phi_{ij}(t)$-style framework, if the relevant coarse-graining and read-out maps are independently specified.

$$\mathcal{B}_1 \xleftrightarrow{\Phi_{12}(t)} \mathcal{B}_2 \xleftrightarrow{\Phi_{23}(t)} \mathcal{B}_3 \quad \cdots \quad \mathcal{B}_n$$

plastic coupling history $K(t,s)$ $\quad\rightsquigarrow\quad$ future synchronization and stable read-out

Figure 3: Schematic only. The figure represents a plastic observable-algebra network. It is not a connectome, a data plot, or a claim of measured neural coupling.

This reading is compatible with ordinary neuroscience vocabulary, but it is not a replacement for it. It does not specify neurotransmitters, spike timing, synaptic weights, or network

physiology. It gives a structural interface in which such details, if independently modeled, could potentially constrain a future construction of $\mathcal{B}_{\text{brain}}$.

Remark 4.1 (Candidate brain algebra). Here $\mathcal{B}_{\text{brain}}$ denotes a full candidate observable algebra for the brain component within the embodied observer regime. If such a $\mathcal{B}_{\text{brain}}$ exists at a chosen coarse-graining scale, it is understood to satisfy $\mathcal{B}_{\text{brain}} \subseteq \mathcal{A}_c$ and $\mathcal{B}_{\text{brain}} \neq \mathcal{A}_c$: the brain component anchors, but does not exhaust, the accessible algebra of the embodied regime. The paper does not construct $\mathcal{B}_{\text{brain}}$ from neuroscience data, and it does not assume that such a construction is unique.

5 Internal Integration Workspace and the Self-Model

The self-model is read here as an internal integration workspace within a single accessible algebra. It is not a new independent algebra outside $\mathcal{A}_c$. It is a stable internal substructure produced by overlap, coupling, and recurrent calibration among multiple read-out subalgebras.

The analogy with Foundation II is partial. Foundation II studies an external shared sector across independent source states under named conditions [2]. Here the internal reading concerns subalgebras and comparison maps within one embodied accessible algebra. The two settings are mechanism-homologous only: no Foundation II datum, theorem, or non-triviality conclusion is inherited here.

The contrast can be stated more explicitly. In the external case, the problem is how multiple source chains or regimes can share a stable overlap while retaining private complements. In the internal case, the problem is how multiple read-out structures inside one embodied accessible algebra can support a stable reportable self-model while retaining local bodily, affective, memory, or pre-report dynamics. The same words "shared" and "private" are used, but the index sets and interpretation differ.

This is why the paper does not simply copy the external construction. The internal construction requires its own datum: read-out subalgebras $\mathcal{B}_i$, conditional expectations E_i, comparison maps R_α, K-dependent coupling maps $\Phi_{ij}^K(t)$, a faithful normal state ω, the tolerance δ, and the report window T. These objects identify the chosen biological coarse-graining scale. Without such a datum, "internal workspace" would remain only a metaphor.

At the internal scale, different read-out subalgebras contribute different aspects of the reportable self-model. Sensorimotor channels stabilize bodily boundary and action affordance. Interoceptive and affective channels stabilize urgency, salience, and valence. Memory channels stabilize temporal continuity. Language channels stabilize reportable narrative.

The reportable self-model is not simply identical with any one of these channels. It is the stable overlap generated by their recurrent coupling. When the overlap is strong and coordinated, the regime reports "I am doing this" or "I decided this." When the overlap weakens or fragments, the reportable structure may become less unified.

This stable overlap is what the paper calls the internal integration workspace $\mathcal{W}_{\text{int}} \equiv \mathcal{W}_{\text{int}}$ at the informal level. Section 6 gives the authoritative formalization; Appendix A only restates its assumptions and consequences.

The construction route is simple but conditional. First choose a scale at which read-out structures can be idealized as subalgebras. Then choose channel expectations, normal read-out comparison maps, and a report-window tolerance. Their exact operator equalizer is always an operator system. It becomes the von Neumann algebra $\mathcal{W}_{\text{int}}$ only under the additional fixed-point hypotheses stated in Section 6. This distinction prevents scalar state agreement from being mistaken for an intrinsic shared algebra.

This route also clarifies what the self-model is not. It is not a point that owns the channels. It is not the language channel alone. It is not the entire accessible algebra. It is a stable readable substructure within $\mathcal{A}_c$, dependent on the selected scale, coupling family, state, and report window.

This does not mean that the self-model is merely an illusion. The self-model is effective within $\mathcal{A}_c$. It can plan, report, repair, remember, and assume responsibility. What is denied is not its effectiveness, but its status as an independent point-subject.

The internal workspace also has a private side. Some bodily, affective, or pre-report dynamics may shape the workspace without becoming fully shared-readable across internal channels or externally reportable to other observers. This is why the paper distinguishes first-person report from the full content of experience.

The relation to Paper I is motivational rather than central. Paper I argues that geometry and modular flow are not independently deformable in the relevant algebraic setting [4]. The present paper uses that lesson only as a caution against separating "objective brain" and "subjective report" into two independently specified domains.

6 Formal Core

Mathematical status of the results. The formal content has four distinct grades. Proposition 6.1 is an operator-algebraic statement about equalizers of named maps. Propositions 6.2 and 6.3 are conditional consequences of explicit interface hypotheses. Proposition 6.5 is a finite Type I model check showing that those hypotheses are jointly satisfiable and non-vacuous. The interpretation of $\mathcal{W}_{\text{int}}$ as a biological self-model remains an interface reading, not a theorem of neuroscience or a consequence imported from Foundation II.

The exact equalizer below is separated from its tolerance halo. A general equalizer of normal unital completely positive maps is a σ-weakly closed operator system, but it need not be an algebra. The von Neumann conclusion therefore uses an explicit automorphic fixed-point hypothesis. Appendix A defers to this section as the authoritative formal statement and records the same assumptions without introducing a parallel construction.

6.1 Setup

An internal read-out datum is a tuple

$$\mathfrak{D}_{\text{int}} = (\mathcal{A}_c, \omega, \{\mathcal{B}_i\}_{i\in I}, \{E_i\}_{i\in I}, \{R_\alpha\}_{\alpha\in J}, \{\Phi_{ij}^K(t)\}_{i,j\in I}, K, \delta, T),$$

where $\mathcal{A}_c$ is an accessible von Neumann algebra, ω is a faithful normal state, $\mathcal{B}_i \subseteq \mathcal{A}_c$ are read-out von Neumann subalgebras, and $E_i : \mathcal{A}_c \to \mathcal{B}_i$ are faithful normal conditional expectations. The maps $R_\alpha : \mathcal{A}_c \to \mathcal{A}_c$ are normal unital completely positive read-out comparison maps, with a distinguished identity map among them. The maps $\Phi_{ij}^K(t) : \mathcal{A}_c \to \mathcal{A}_c$ are normal unital completely positive coupling maps whose declared time dependence is parameterized by the history object K. Finally, $T > 0$ is the report-window length and $\delta \geq 0$ is the operator-norm agreement tolerance.

Neither δ nor any other component of the datum is inferred from neuroscience data here. At $\delta = 0$ agreement is exact. For $\delta > 0$, define only the unit-ball tolerance locus

$$\mathcal{W}_{\text{int}}^{(\delta)} = \left\{ a \in \text{Ball}(\mathcal{A}_c) : \sup_{\alpha,\beta\in J} \|R_\alpha(a) - R_\beta(a)\| \leq \delta \right\}.$$

The unit-ball restriction prevents a tolerance statement from being defeated by scalar rescaling. The exact agreement locus is

$$\mathrm{Eq}(R) = \{a \in \mathcal{A}_c : R_\alpha(a) = R_\beta(a) \text{ for all } \alpha, \beta \in J\},$$

and the symbol $\mathcal{W}_{\text{int}}$ is reserved for this exact locus when the conditions below make it a non-trivial von Neumann algebra. The approximate set $\mathcal{W}_{\text{int}}^{(\delta)}$ is not asserted to be linear or multiplicatively closed.

Proposition 6.1 (Read-out equalizer)**.** *For any family of normal unital completely positive maps $\{R_\alpha\}_{\alpha\in J}$, the exact equalizer* $\mathrm{Eq}(R)$ *is a σ-weakly closed operator system. If instead $R_g = \alpha_g$ is a point–σ-weakly continuous action of a finite or compact group G by normal $*$-automorphisms, with $R_e = \mathrm{id}$, then*

$$\mathrm{Eq}(R) = \mathcal{A}_c^G$$

is a von Neumann subalgebra. Haar averaging gives a normal faithful conditional expectation

$$E_{\text{int}}(a) = \int_G \alpha_g(a)\, d\mu(g) \quad \textit{with} \quad \mathrm{Ran}(E_{\text{int}}) = \mathcal{A}_c^G.$$

If $\omega \circ \alpha_g = \omega$ for all g, this expectation is ω-preserving.

Proof. The equalizer is the intersection of the kernels of the normal linear maps $R_\alpha - R_\beta$, hence is σ-weakly closed. Unital positive maps preserve adjoints, so the equalizer contains the identity and is self-adjoint and linear. It is therefore an operator system. For an automorphic group action, the equalizer is the fixed-point set $\mathcal{A}_c^G$, which is also closed under products and is consequently a von Neumann subalgebra. The normalized group average is normal, faithful, unital, positive, idempotent, and bimodular over $\mathcal{A}_c^G$, giving the claimed conditional expectation and range equality. □

The proposition deliberately does not promote an arbitrary CP-map equalizer to an algebra. The automorphic calibration hypothesis is a model choice that is satisfied by the finite witness below; other mechanisms for multiplicative closure remain open.

6.2 Conditional Hull Construction

Assume the following conditions:

(H0) **Channel selection.** The maps $E_i : \mathcal{A}_c \to \mathcal{B}_i$ are faithful normal conditional expectations onto the named read-out subalgebras.

(H1) **Automorphic comparison.** The comparison maps are a finite or compact normal automorphic action $R_g = \alpha_g$, and ω is invariant under that action.

(H2) **Exact common range and non-triviality.** The fixed-point equalizer obeys

$$\mathbb{C}\mathbf{1} \subsetneq \mathcal{W}_{\text{int}} := \mathcal{A}_c^G = \bigcap_{i\in I} \mathcal{B}_i \subsetneq \mathcal{A}_c.$$

This equality is a substantive interface hypothesis; scalar equality of state expectations would not imply it.

(H3) **History-dependent finite-window stability.** The declared coupling family depends on K and satisfies, for $0 \le t \le T$,

$$\Phi_{ij}^{K}(t)\big(\mathrm{Ball}(\mathcal{W}_{\mathrm{int}})\big) \subseteq \mathcal{W}_{\mathrm{int}}^{(\delta)} \qquad (i, j \in I).$$

At $\delta = 0$ this is exact invariance. Positive δ records an approximate report regime and does not define a new algebra.

Interpretation of the hypotheses. Conditions (H0)–(H2) state exactly where algebraic closure and a common channel range enter. They are not neuroscience conclusions. Condition (H3) is operational and finite-window: it uses the named tolerance, coupling family, and history object rather than an undefined phrase "up to the regime tolerance."

Hypothesis	Formal burden	Interface meaning
(H0)	Existence of faithful normal conditional expectations onto selected subalgebras.	Channel-readable content can be selected at the chosen scale.
(H1)	Automorphic read-out comparisons with an invariant faithful state.	Exact agreement has a specified algebraic mechanism.
(H2)	The fixed algebra equals the intersection of channel algebras and is proper and non-scalar.	The workspace is common to all channels but does not exhaust the accessible algebra.
(H3)	K-parameterized coupling sends the workspace unit ball into the δ-locus over $[0, T]$.	The candidate remains readable over the declared report window.

Table 4: Interpretation of the formal hypotheses. The right column is only an interface reading, not an empirical derivation of the hypotheses.

Proposition 6.2 (Conditional Hull Construction)**.** *Fix an internal read-out datum $\mathfrak{D}_{\mathrm{int}}$ satisfying (H0)–(H3). Then the exact read-out equalizer is the non-trivial proper von Neumann algebra*

$$\mathcal{W}_{\mathrm{int}} = \mathrm{Eq}(R) = \bigcap_{i \in I} \mathcal{B}_i.$$

The group average in Proposition 6.1 is a faithful normal ω-preserving conditional expectation $E_{\mathrm{int}} : \mathcal{A}_c \to \mathcal{W}_{\mathrm{int}}$, so the equality $\mathrm{Ran}(E_{\mathrm{int}}) = \mathcal{W}_{\mathrm{int}}$ is a consequence of the stated hypotheses rather than an additional stipulation. Condition (H3) gives δ-stability over the report window, with exact invariance when $\delta = 0$.

Proof. Proposition 6.1 and (H1) identify the equalizer with a fixed-point von Neumann algebra and construct E_{int}. Hypothesis (H2) identifies that algebra with the exact intersection of the channel ranges and supplies proper non-triviality. Hypothesis (H3) is precisely the asserted finite-window stability statement. □

6.3 Conditional Common/Private Read-Out Decomposition

Proposition 6.3 (Conditional Common/Private Decomposition)**.** *Suppose Proposition 6.2 holds. Then the common channel algebra is exactly*

$$\bigcap_{i \in I} \mathcal{B}_i = \mathcal{W}_{\mathrm{int}},$$

so for every channel

$$\mathcal{B}_i^{\text{com}} := \mathcal{B}_i \cap \mathcal{W}_{\text{int}} = \mathcal{W}_{\text{int}}.$$

and each channel has the exact linear splitting

$$\mathcal{B}_i = \mathcal{W}_{\text{int}} \oplus \ker(E_{\text{int}}|_{\mathcal{B}_i}).$$

The sum is a direct sum of linear spaces, not generally a direct sum of algebras. If, in addition, there are von Neumann subalgebras $\mathcal{P}_i \subseteq \mathcal{B}_i$ satisfying

$$\mathcal{B}_i = \mathcal{W}_{\text{int}} \vee \mathcal{P}_i, \qquad \mathcal{P}_i \cap \mathcal{W}_{\text{int}} = \mathbb{C}\mathbf{1}, \tag{P}$$

then $\mathcal{P}_i$ is an exact private algebra relative to the common workspace. The existence of such $\mathcal{P}_i$ is an explicit extra hypothesis in general and is verified in Proposition 6.5.

Proof. The common-algebra equality is (H2). Since $E_{\text{int}}|_{\mathcal{W}_{\text{int}}} = \text{id}$, every $b \in \mathcal{B}_i$ has the unique linear decomposition

$$b = E_{\text{int}}(b) + \big(b - E_{\text{int}}(b)\big),$$

with the first term in $\mathcal{W}_{\text{int}}$ and the second in the restricted kernel. Their intersection is $\{0\}$. Under (P), the displayed join and scalar intersection give the stronger common/private algebra statement by definition of generated von Neumann algebra. □

Remark 6.4 (General private-algebra problem)*.* The linear kernel splitting is automatic under the conditional expectation constructed above. A private von Neumann algebra satisfying (P) is not automatic and remains open for general read-out data. The finite witness below proves only that the stronger package is inhabited.

Why this decomposition matters. The common/private distinction is not added for mathematical ornament. It is the formal place where the paper separates reportable self-model from the full range of processes shaping experience and behavior. A memory trace, bodily tension, affective bias, or subreport motor tendency may influence the self-model without being shared-readable in the selected hull. Conversely, what enters $\mathcal{B}_i^{\text{com}}$ is available to the internal overlap relative to the chosen E_{int}.

This gives a precise sense in which the self-model can be effective without being exhaustive. It can guide report and action because it is a stable shared-readable structure. It can fail to capture everything because each channel may retain a private complement. The framework therefore avoids both extremes: it does not treat the self-model as a fiction, and it does not treat the self-model as the whole mind.

6.4 Finite-Dimensional Shared-Factor Witness

Proposition 6.5 (Finite shared-factor witness)**.** *There is a finite Type I internal read-out datum satisfying (H0)–(H3) at $\delta = 0$, with non-trivial proper workspace $\mathcal{W}_{\text{int}}$, exact common channel algebra, and non-zero private algebras satisfying (P).*

Proof. Let

$$\mathcal{A}_c = M_2(S) \otimes M_2(P_1) \otimes M_2(P_2)$$

and let $\omega = \tau_S \otimes \tau_1 \otimes \tau_2$, where each τ is the normalized trace. Define two channel algebras

$$\mathcal{B}_1 = M_2(S) \otimes M_2(P_1) \otimes \mathbb{C}\mathbf{1}, \qquad \mathcal{B}_2 = M_2(S) \otimes \mathbb{C}\mathbf{1} \otimes M_2(P_2),$$

with the trace-preserving conditional expectations

$$E_1(a_S \otimes a_1 \otimes a_2) = a_S \otimes a_1 \otimes \tau_2(a_2)\mathbf{1}, \qquad E_2(a_S \otimes a_1 \otimes a_2) = a_S \otimes \tau_1(a_1)\mathbf{1} \otimes a_2,$$

extended linearly. These maps are faithful and normal, so (H0) holds.

Let G be the finite group generated by independent Pauli conjugations on P_1 and P_2, with phases factored out, and set

$$R_{(g_1,g_2)} = \mathrm{Ad}_{\mathbf{1}_S \otimes g_1 \otimes g_2}\,.$$

The only matrices fixed by all Pauli conjugations on either private qubit are scalars. Hence

$$\mathrm{Eq}(R) = \mathcal{A}_c^G = M_2(S) \otimes \mathbb{C}\mathbf{1} \otimes \mathbb{C}\mathbf{1} =: \mathcal{W}_{\text{int}}.$$

This algebra is neither $\mathbb{C}\mathbf{1}$ nor $\mathcal{A}_c$, and direct tensor-factor inspection gives

$$\mathcal{B}_1 \cap \mathcal{B}_2 = \mathcal{W}_{\text{int}}.$$

The trace state is G-invariant. The group average is therefore the explicit expectation

$$E_{\text{int}}(a_S \otimes a_1 \otimes a_2) = a_S \otimes \tau_1(a_1)\mathbf{1} \otimes \tau_2(a_2)\mathbf{1}$$

on simple tensors, extended linearly, with range $\mathcal{W}_{\text{int}}$, verifying (H1)–(H2).

To use the history datum non-trivially, take

$$K(t,s) = \kappa e^{-\lambda(t-s)}\mathbf{1}_{0\le s\le t}, \qquad c_K(t) = \int_0^t K(t,s)\,ds,$$

where $\kappa, \lambda > 0$, and set

$$U_K(t) = \exp\big[-ic_K(t)\big(Z_{P_1} \otimes Z_{P_2}\big)\big]\,.$$

Define $\Phi_{12}^K(t) = \mathrm{Ad}_{\mathbf{1}_S \otimes U_K(t)}$ and $\Phi_{21}^K(t) = \mathrm{Ad}_{\mathbf{1}_S \otimes U_K(t)^*}$, with $\Phi_{11}^K(t) = \Phi_{22}^K(t) = \mathrm{id}$. The cross-channel maps are non-trivial normal automorphisms on the private factors: the $Z_{P_1} \otimes Z_{P_2}$ interaction acts on both named channel factors rather than serving as unused datum. All four maps fix $\mathcal{W}_{\text{int}}$ pointwise, establishing (H3) with $\delta = 0$ for every finite report window.

Finally set

$$\mathcal{P}_1 = \mathbb{C}\mathbf{1}_S \otimes M_2(P_1) \otimes \mathbb{C}\mathbf{1}, \qquad \mathcal{P}_2 = \mathbb{C}\mathbf{1}_S \otimes \mathbb{C}\mathbf{1} \otimes M_2(P_2).$$

Then $\mathcal{B}_i = \mathcal{W}_{\text{int}} \vee \mathcal{P}_i$ and $\mathcal{P}_i \cap \mathcal{W}_{\text{int}} = \mathbb{C}\mathbf{1}$. Both private algebras are non-zero; for example $\mathbf{1}_S \otimes X_{P_1} \otimes \mathbf{1} \in \mathcal{P}_1$ is non-scalar. Thus the complete formal package is jointly satisfiable without importing a Foundation II theorem. □

6.5 Failure Modes

The same formal setup also identifies ways in which the construction may fail. These failure modes are important because they prevent the paper from treating a unified self-model as automatic.

(M1) The selected read-out structures may not admit the required conditional expectations. In that case the chosen scale is not suitable for this construction.

(M2) A general CP-map equalizer may fail to be multiplicatively closed, or the exact equalizer may fail to equal the intersection of channel algebras. In either case no von Neumann workspace follows from the present proposition.

(M3) The exact equalizer may be only $\mathbb{C}\mathbf{1}$, or may be all of $\mathcal{A}_c$. Either case defeats the intended non-trivial internal workspace reading.

(M4) The coupling maps may fail the explicit δ-stability bound over $[0, T]$. In that case the workspace is too unstable for the declared report regime.

(M5) Private algebras $\mathcal{P}_i$ satisfying (P) may not exist even when the linear kernel splitting does. In that case exact private-algebra language remains open.

These failure modes are not pathologies in the clinical sense. They are formal warnings. A future empirical or mathematical application would need to show not only that a datum can be written down, but that the chosen datum avoids these failure modes in the regime under study.

6.6 Fragmentation Criterion

The internal workspace may fragment when the conditions supporting stable overlap fail. A formal fragmentation regime may be indicated by one or more of the following:

(F1) The coupling graph on I disconnects into weakly coupled or uncoupled components.

(F2) For relevant comparison maps and some workspace unit-ball element, the disagreement exceeds the declared δ over the report window.

(F3) The exact equalizer ceases to coincide with the intersection of the relevant channel algebras.

(F4) The K-parameterized coupling family fails to preserve even the approximate locus $\mathcal{W}_{\text{int}}^{(\delta)}$ over $[0, T]$.

Under such conditions, multiple local self-model candidates may become structurally possible. This is a formal criterion, not a diagnosis. It does not identify DID, split-brain, or any clinical state.

7 First-Person Report and Timing

First-person report is treated as a late, reportable phase of a broader internal stabilization process. This section uses Libet-style timing only as a compatible re-reading. It is not evidence for the framework, and it is not a general theory of decision.

In standard presentations of Libet-style wrist-flexion experiments, a readiness potential is reported before the subject's reported awareness of deciding, and both precede the action [8, 29, 30]. The timing values usually associated with this paradigm are approximate and paradigm-dependent. They are drawn from self-paced, low-stakes actions, and extrapolation to complex or consequential decisions is not supported by the present framework. Schurger et al.'s stochastic-accumulator account undercuts the reading of the readiness potential as an already settled decision: time-locking stochastic fluctuations can generate the averaged signal. The timing table below is therefore a compatible illustration only and carries no confirmatory, causal, or predictive weight.

The readiness potential is treated here as a window into action-preparation dynamics, not as a direct marker of completed decision. In the present language, it can be read as an early

measurable trace of action-candidate stabilization. It does not show that the internal integration-workspace model is true.

Timing	**Compatible structural re-reading**	**Observable marker**
Approx. -550 ms	Action-candidate stabilization begins to become measurable in the relevant motor-preparation regime.	Readiness potential.
Approx. -200 ms	Reportable self-attribution becomes available: "I decided."	W time report.
Action time	Motor output occurs.	Movement.

Table 5: Schematic timing diagram. It is not an empirical data plot and does not generalize to complex or consequential decisions.

The compatible re-reading is that the reportable "I decided" is not the first event in the action process. It belongs to the stage at which an action tendency has become stable enough to be attributed to the self-model. This does not make first-person report worthless. It locates it in the reportable phase of a longer internal process.

This also blocks a misleading opposition. It is not "the brain decided and I was bypassed." The internal workspace is stabilized within the person-level regime through which action is prepared, attributed, and executed. What is misleading is the identification of the person with the late linguistic report alone.

Responsibility is therefore not erased. The relevant responsible structure is the embodied regime with its history-dependent couplings, plasticity, repair capacities, and future modifiability. Responsibility is not located in a single sentence, "I decided," but in a trainable and accountable condition-chain.

8 Neuroscience Interfaces

This section gives interface points only. It is not a review article, and it does not claim to replace the theories or clinical categories it mentions. The cases below are selective examples rather than a complete inventory. For all empirical phenomena discussed below, the phenomenon is used as compatible re-reading only, not as evidence for the framework. The neuroscience citations in this section are orientation markers: they locate where the algebraic language might touch existing empirical or theoretical vocabularies. They are not evidential premises for the formal results in Section 6, and they do not make the framework empirically confirmed. External references in this section require author verification before submission.

The table is included to prevent a misleading impression. The paper is not collecting neuroscientific evidence for the algebraic framework. Instead, it identifies where existing empirical or theoretical vocabularies would touch the algebraic objects introduced above: coupling maps, report boundaries, external calibration, private complements, and hull fragmentation. A reader should therefore treat the table as an interface map, not as an argument from neuroscience to the framework.

Interface	External theory or phenomenon	Structural reading in this framework
Split-brain	Split-brain cases involve disrupted interhemispheric communication [9, 10].	Can be read as coupling disconnection and possible internal-hull fragmentation.
DID	DID is a clinically complex dissociative condition and should not be diagnosed from a formal model [11, 12].	Can be read as protected decoupling among memory, affective, bodily, and narrative channels.
Sleep	Sleep involves reduced sensory processing and motor output with memory and network maintenance [13, 14].	Can be read as offline reweighting of $K(t, s)$ under reduced immediate action demand.
Dreaming	Dreaming involves reduced external input with internally active memory, affective, and imagery systems [15, 16].	Can be read as weakened external calibration while internal-hull stabilization continues.
Meditation	Meditation research studies training of attention, interoception, and affective regulation [17, 18].	Can be read as modulation of attention and the report boundary.
OBE / altered self-location	OBE and dissociative states involve altered body-ownership and self-location [19, 20].	Can be read as body-anchoring and self-location decoupling or remapping.
GWT	Global Workspace Theory describes consciousness in terms of global availability [21, 22].	Can be read as a reportable phase of internal-hull stabilization.
Predictive processing	Predictive processing describes perception and action through prediction and error dynamics [23, 24].	Can be read as one class of coupling update within the embodied observable-algebra network.
HOT	Higher-order theories locate awareness in states about other states [25, 26].	Can be read as one read-out structure making another available at the report boundary.

Table 6: Selective neuroscience and consciousness-theory interfaces. Each structural reading is a compatible re-reading only, not evidence for the framework and not a replacement for the external theory.

The table also explains why the section does not expand every row into a separate literature review. The most important rows for the formal core are split-brain, global availability, and sleep/dreaming. Split-brain interfaces with fragmentation, global availability interfaces with reportable stabilization, and sleep/dreaming interfaces with history-dependent coupling and external calibration. The remaining rows are included to show where the same language may connect later, not to claim coverage of clinical or contemplative phenomena. None of the structural readings in the table replaces the external theory or clinical category it interfaces with.

Split-brain and fragmentation. Split-brain cases provide the cleanest interface to the formal fragmentation criterion in Section 6. When a major coupling route is disrupted, some read-out subfamilies may no longer support one unified candidate workspace. This does not mean that the formal model diagnoses split-brain patients. It only says that coupling disconnection can be read as one biological interface to workspace fragmentation.

Global availability and report. Global Workspace Theory is the most direct theory-level interface because it focuses on global availability across multiple systems. In the present framework, global availability can be read as a reportable phase of internal-hull stabilization across read-out structures. The paper does not replace GWT, does not derive its architecture, and does not claim that GWT is wrong.

Sleep, dreaming, and history-dependent coupling. Sleep and dreaming provide the most direct interface to $K(t, s)$ and external calibration. Sleep can be read as a state in which immediate external read-out and action demand are reduced while coupling history is reweighted; dreaming can be read as a state of altered external calibration in which transient world-model and self-model structure may continue without direct external anchoring. The paper does not offer a complete theory of sleep or dreaming and does not treat dream content as evidence for the framework.

First-person timing as a restricted interface. The Libet-style discussion in Section 7 is included because it gives a restricted timing case for first-person report. Readiness potential is read as a measurable trace of action-preparation dynamics, not as a direct marker of completed decision. This interface is limited to self-paced, low-stakes action paradigms and is not generalized to complex or consequential decisions.

9 Guardrails and Limitations

9.1 Guardrails

1. Not a quantum-brain hypothesis. This paper is not a quantum-brain hypothesis. The algebraic reading is allowed to operate at an effective or classical-observable level, and it does not require long-lived quantum coherence in the brain. The counter-claim not made is: "the brain is a quantum computer that produces consciousness."

2. Not an empirical derivation of a testable construction of $\mathcal{B}_{\text{brain}}$. This paper is not an empirical derivation of a testable construction of $\mathcal{B}_{\text{brain}}$. It introduces a structural interface that future work could potentially constrain using neuroscience data. The counter-claim not made is: "neuroscience data already determine the observable algebra of the brain."

3. Not exhaustive over psychological functions. This paper is not exhaustive over psychological functions, including qualia, emotion, imagery, clinical symptoms, or all cognition. It describes one structural reading of internal stabilization and first-person report. The counter-claim not made is: "all mental functions are explained by the plastic observable-algebra network."

4. Not a proof of qualia or a solution to the hard problem. This paper is not a proof of qualia or a solution to the hard problem. It does not claim to solve the hard problem of consciousness, and it does not reduce first-person experience to third-person description. The counter-claim not made is: "the hard problem has been solved by neural algebra."

5. Not authorization for consciousness migration. This paper is not authorization for consciousness migration, digital immortality, or first-person transfer. Functional copying, if discussed elsewhere, does not establish identity transfer. The counter-claim not made is: "a copy of the network is the same first-person subject."

6. Not authorization for brain-computer-interface mind reading. This paper does not claim that brain-computer interfaces can read, write, or transfer first-person consciousness. BCI results can be read as partial access to selected sensory, motor, or language-output channels, not as direct access to the full internal workspace. The counter-claim not made is: "recording enough neural activity would read the mind or capture the subject."

7. Not a claim about the soul, near-death experience, or post-mortem continuity. This paper does not prove, disprove, or redescribe any metaphysical doctrine of the soul. Phenomena such as out-of-body experience are treated here only as altered self-location readings within the framework, not as evidence for or against first-person persistence beyond the embodied regime. The counter-claim not made is: "the soul exists as a separable entity the framework endorses."

9.2 Limitations

First, the internal workspace candidate depends on the selected coarse-graining scale and on the choice of coupling family. Different choices may produce different workspace candidates. The paper does not resolve this scale- and choice-dependence.

Second, cross-scale consistency remains coarse. Synaptic plasticity may operate over milliseconds, seconds, hours, days, and longer timescales, while first-person report timing may be measured in a different regime. This paper does not provide a full multi-scale neural dynamics model.

Third, the physical and neurobiological plausibility of assumptions (H1)–(H3) in Section 6 and Appendix A requires separate work. They are explicit modeling assumptions, not conclusions derived from neurophysiology.

Fourth, the relation between one embodied observer regime and the external shared sector H_{12} from Foundation II is not covered here. The paper is about the internal anchor of one regime. It does not construct a full bridge from $\mathcal{W}_{\text{int}}$ to social or intersubjective worlds, and it imports no Foundation II theorem.

10 Research Program

The framework suggests a conservative research program. It does not state that existing data validate the model. It asks how future data could potentially constrain pieces of the proposed interface.

Connectome data could potentially constrain candidate coupling topologies $\Phi_{ij}^{K}(t)$. Such data would not by themselves identify $\mathcal{B}_{\text{brain}}$, but they may provide empirical pressure on which coarse-grained read-out units are plausible. Any such use would require independent modeling choices.

EEG and fMRI signatures could potentially constrain large-scale integration and fragmentation regimes. They would not directly measure the internal integration workspace $\mathcal{W}_{\text{int}}$. They might identify regimes in which a simplified coupling model becomes more or less plausible.

Perturbation data could be used to test simplified cases of hull stability. Stimulation, lesion, or pharmacological studies may create natural interface cases. This paper does not propose a protocol and does not treat such data as confirmation.

Plasticity models could provide coarse constraints on $K(t, s)$-like history dependence. The relevant question would be how prior coupling history changes future readability and synchronization. This remains a research program rather than a result.

Long-term potentiation and related synaptic plasticity mechanisms could potentially provide one empirical interface for this question [27, 28]. Such mechanisms are compatible with reading memory as coupling history, but the present framework does not claim that any single plasticity mechanism implements $K(t, s)$. They would at most constrain simplified biological cases.

Formal direction. The most direct formal next step is to constrain or estimate the explicit operator-norm tolerance δ and report window T in a named biological model. One could also compare the present norm criterion with state-distance or operational distinguishability criteria. This would test how robust the exact workspace and its tolerance locus are under a change of metric rather than leaving tolerance undefined.

A second formal direction is to study uniqueness and functoriality under coarse-graining. The present paper explicitly allows different choices of $\{\mathcal{B}_i\}$, E_i, R_α, $\Phi^K_{ij}(t)$, δ, and T to produce different workspace candidates. Later work could ask whether some candidates are stable under refinement or coarsening of the biological scale. Such a result would be much stronger than anything claimed here.

Empirical direction. The empirical direction is not to look for $\mathcal{W}_{\text{int}}$ directly. The internal workspace is not an observable brain signal. A more conservative program would seek constraints on the ingredients of the datum: candidate read-out families, coupling topologies, perturbation sensitivity, and history-dependent plasticity. If these ingredients can be bounded in simplified settings, the formal hypotheses (H0)–(H3) could become less arbitrary.

This still would not validate the full framework. It would only show that some idealized datum is plausible for some limited biological regime. That is the right scale of empirical ambition for an anchoring interface paper.

11 Conclusion

The brain-centered embodied observer regime can be read as a biological anchor for an accessible algebra. Under hypotheses (H0)–(H3), a plastic observable-algebra network has the exact internal fixed-point workspace $\mathcal{W}_{\text{int}}$, which may be read at the interface level as supporting first-person report. The finite-dimensional witness (Proposition 6.5) confirms non-triviality of $\mathcal{W}_{\text{int}}$ in the Pauli-qubit model; the general private-complement condition (P) remains open.

The workspace $\mathcal{W}_{\text{int}}$ plays a role analogous to the external shared sector H_{12} in Foundation II, but no equality, functor, or theorem-level inheritance is asserted (see the typed hull namespace definition above). This conditional interface does not solve consciousness, reduce selfhood to neural machinery, or prove qualia.

A Optional Formal Skeleton for the Internal Integration Workspace

This appendix is subordinate to the Formal Core in Section 6. It repeats the datum and the two conditional consequences for reference, but it does not define a second overlap construction. If wording here differs from Section 6, the definitions and hypotheses in Section 6 govern. In particular, the workspace is the exact read-out equalizer $\mathcal{W}_{\text{int}}$, the positive-δ object is only a unit-ball tolerance locus, and the conditional expectation is obtained from the fixed-point action rather than stipulated independently.

A.1 Provisional Definition: Internal Read-Out Datum

Definition A.1 (Appendix restatement of the internal read-out datum)**.** The authoritative internal read-out datum of Section 6 is

$$\mathfrak{D}_{\text{int}} = (\mathcal{A}_c, \omega, \{\mathcal{B}_i\}_{i\in I}, \{E_i\}_{i\in I}, \{R_\alpha\}_{\alpha\in J}, \{\Phi_{ij}^K(t)\}_{i,j\in I}, K, \delta, T).$$

Here $\delta \geq 0$ is a declared operator-norm tolerance on the unit ball over the report window $[0, T]$; it is a modeling parameter, not a quantity inferred from neuroscience. The comparison maps R_α and coupling maps $\Phi_{ij}^K(t)$ have the distinct typed roles specified in the Setup.

Remark A.2 (Scale-dependence)*.* The cardinality and identity of $\{\mathcal{B}_i\}_{i\in I}$ and the existence of the maps $\{E_i\}$ depend on the coarse-graining scale chosen within the embodied regime. Proposition A.4 is understood at a fixed such scale and report window, not as an invariance claim across scales.

Remark A.3 (Exact and approximate objects)*.* At $\delta = 0$, $\mathcal{W}_{\text{int}} = \text{Eq}(R)$ is the exact agreement locus. At $\delta > 0$, $\mathcal{W}_{\text{int}}^{(\delta)}$ is the operator-norm tolerance locus defined in Section 6. No algebraic closure is claimed for $\mathcal{W}_{\text{int}}^{(\delta)}$. This explicit parameter replaces the former undefined phrase "up to the regime tolerance."

A.2 Appendix Restatement of the Conditional Construction

Proposition A.4 (Appendix restatement of the internal workspace)**.** *Under the same (H0)–(H3) as Proposition 6.2,*

$$\mathcal{W}_{\text{int}} = \text{Eq}(R) = \mathcal{A}_c^G = \bigcap_{i\in I} \mathcal{B}_i$$

is a non-trivial proper von Neumann algebra. The group average is the faithful normal ω-preserving expectation $E_{\text{int}} : \mathcal{A}_c \to \mathcal{W}_{\text{int}}$, and $\Phi_{ij}^K(t)(\text{Ball}(\mathcal{W}_{\text{int}})) \subseteq \mathcal{W}_{\text{int}}^{(\delta)}$ for $0 \leq t \leq T$.

Proof. This is Proposition 6.2, restated without altering its hypotheses. Proposition 6.1 supplies the fixed-point algebra and conditional expectation; (H2) and (H3) supply the exact channel intersection and tolerance stability. □

Remark A.5 (What the construction does not show)*.* Proposition A.4 does not prove that the hypotheses hold in biological brains. It does not give an empirical construction of $\mathcal{B}_{\text{brain}}$. It also does not prove uniqueness of $\mathcal{W}_{\text{int}}$, cross-scale invariance, or first-person identity transfer. The automorphic comparison action, exact channel intersection, and finite-window bound remain explicit interface hypotheses.

Remark A.6 (Relation to Proposition 6.2). Proposition A.4 is not an independent result. Section 6 is authoritative. The finite model in Proposition 6.5 verifies the same hypotheses and also verifies the stronger private-algebra condition (P).

A.3 Conditional Common / Private Decomposition

Proposition A.7 (Conditional common/private decomposition)**.** *Under Proposition A.4,*

$$\bigcap_{i\in I} \mathcal{B}_i = \mathcal{W}_{\text{int}}$$

and, for every channel,

$$\mathcal{B}_i = \mathcal{W}_{\text{int}} \oplus \ker(E_{\text{int}}|_{\mathcal{B}_i})$$

as linear spaces. A private von Neumann algebra $\mathcal{P}_i$ with $\mathcal{B}_i = \mathcal{W}_{\text{int}} \vee \mathcal{P}_i$ and $\mathcal{P}_i \cap \mathcal{W}_{\text{int}} = \mathbb{C}\mathbf{1}$ requires the additional condition (P) from Proposition 6.3.

Proof. This is the linear and algebraic distinction proved in Proposition 6.3. The finite witness Proposition 6.5 realizes (P) with two non-zero private qubit algebras. No Foundation II result is used. □

A.4 Criterion: Fragmentation Regime

The internal workspace may fragment when the conditions supporting exact or approximate agreement fail. A formal fragmentation regime may be indicated by one or more of the following:

(F1) The coupling graph on I disconnects into weakly coupled or uncoupled components.

(F2) Read-out disagreement exceeds δ on the workspace unit ball over the report window.

(F3) The exact equalizer no longer equals the intersection of the relevant channel algebras.

(F4) The K-dependent coupling maps fail to preserve the δ-locus over $[0, T]$.

Under such conditions, multiple local self-model candidates may become structurally possible. This is a formal criterion, not a diagnosis. It does not identify DID, split-brain, or any clinical state.

The criterion is intentionally weak. It describes a regime in which the single-workspace reading may fail or require refinement. It does not classify all possible fragmentation phenomena and does not replace empirical or clinical analysis.

A.5 Remarks on Limitations of the Skeleton

The construction in Appendix A is a restatement, not a second theorem. It is fixed-scale and does not claim invariance across biological coarse-grainings. A different scale may produce different channel algebras, comparison maps, tolerance, and workspace.

The common/private decomposition and fragmentation criterion are conditional formal tools. The expectation follows from the stated fixed-point action, whereas private algebras satisfying (P) remain an additional general hypothesis. The tools are not presented as completed biological or clinical classifications.

The skeleton does not derive an empirically testable construction of $\mathcal{B}_{\text{brain}}$ from neuroscience data. It does not prove qualia. It does not support first-person identity transfer. It is a constrained formal appendix for an anchoring interface paper.

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